CHASING CONSTANTS, REDUX: REFLECTION AND WARPED METRICS

Abstract. An influential work of Eberle (PTRF 2016) shows how to study the contraction properties of Itô diffusions by means of the Kendall–Cranston "reflection coupling". The present note revisits this work and shows how some aspects of the metric construction can be simplified and optimised.

1. Introduction and the reflection coupling

In the study of Markov processes, it is of substantial interest to study the rates at which processes converge towards their equilibrium measure. A popular strategy to this end is coupling methods, wherein one shows that for two separately-initialised copies of the process in question, one can drive the two processes with a shared source of randomness so that they typically come closer together, as measured by some suitable metric on the ambient space.

A popular test-bed for such strategies is Itô diffusions, with an archetypal class of examples taking the form

$$\mathrm{d}X_t = b\left(X_t\right)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t$$

on $\mathbf{R}^d$ for some suitably-regular drift function b. Supposing that for some $\kappa > 0$, it holds globally that

$$\langle b\left(x\right) - b\left(y\right), x - y\rangle \leq -\kappa \cdot \|x-y\|^2,$$

one sees that the coupling approach works rather straightforwardly: having initialised two copies of the process at x, y respectively, drive them with the same Brownian motion, and observe the almost-sure contraction $\|X_t - Y_t\| \leq \exp\left(-\kappa \cdot t\right) \cdot \|x-y\|$.

In many cases, such a monotonicity may not hold globally, but only for pairs x, y which are sufficiently well-separated in space. To address this setting, other couplings are required; a particularly successful example is the so-called 'reflection coupling'. This coupling is designed to make the noise act directly on the separation of two copies of the diffusion. Starting from distinct points x and y, let $e_t = \frac{X_t - Y_t}{\|X_t - Y_t\|}$ and define $\tau = \inf\left\{t \geq 0 : X_t = Y_t\right\}$. Up to time τ, define the coupled processes by

$$\begin{aligned}
\mathrm{d}X_t &= b\left(X_t\right)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t, \\
\mathrm{d}Y_t &= b\left(Y_t\right)\,\mathrm{d}t + \sqrt{2}\big(I - 2e_t \otimes e_t\big)\,\mathrm{d}B_t.
\end{aligned}$$

After τ, the two processes are run synchronously, so that they remain equal. The matrix $I - 2e_t \otimes e_t$ is the orthogonal reflection in the hyperplane perpendicular to e_t. Consequently, the martingale driving Y_t is again a Brownian motion, and each component of the coupling has the correct marginal law.

Writing $Z_t = X_t - Y_t$ and $\beta_t = \int_0^t \langle e_s, \mathrm{d}B_s\rangle$, the difference process satisfies, before τ

$$\mathrm{d}Z_t = \left(b\left(X_t\right) - b\left(Y_t\right)\right)\,\mathrm{d}t + 2\sqrt{2}\cdot e_t\,\mathrm{d}\beta_t.$$

All stochastic forcing is therefore aligned with this separation. In particular, for $R_t = \|Z_t\|$ there is no transverse Itô correction, and

$$\mathrm{d}R_t = \langle b\left(X_t\right) - b\left(Y_t\right), e_t\rangle\,\mathrm{d}t + 2\sqrt{2}\,\mathrm{d}\beta_t.$$

The d-dimensional coupling problem has thus been reduced to a one-dimensional diffusion inequality for the separation.

Rather than work with the Euclidean separation itself, one introduces a warped distance $\mathsf{d}_f(x,y) = f\left(\|x-y\|\right)$ where f is increasing and concave. Concavity has two roles: it ensures that $f \circ \|\cdot\|$ remains a genuine metric, and its non-positive second derivative provides a negative contribution under reflection coupling. The metric-design problem is consequently to distribute this concavity so that it compensates for regions in which the drift is not contractive, without distorting the metric more than necessary.

The first part of the note gives an explicit one-parameter family of such metrics and an analytic optimization of its parameter. The second part interprets the radial design problem as a one-dimensional control problem with an obstacle–Riccati structure. A simple one-switch specialization then yields a closed-form improvement for a two-level curvature profile.

2. Standing assumptions and notation

Consider the $\mathbf{R}^d$-valued diffusion process $\mathrm{d}X_t = b(X_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t$, and assume that the drift b is sufficiently regular for the diffusion and its reflection coupling to be non-explosive. Define the separation-curvature profile of b for positive r by

$$\kappa_b(r) = \inf\left\{-\frac{\langle b(x) - b(y), x-y\rangle}{\|x-y\|^2} : \|x-y\| = r\right\},$$

so that

$$\langle b(x) - b(y), x-y\rangle \leq -\kappa_b(\|x-y\|)\cdot\|x-y\|^2.$$

Let $\mathcal{F}$ denote the class of functions $f : [0,\infty) \to [0,\infty)$ such that $f(0) = 0$, $f'(0) = 1$, $f' > 0$, and $f'' \leq 0$, again with sufficient regularity for the coupled-generator argument which follows. An important feature of the class $\mathcal{F}$ is that for $f \in \mathcal{F}$, the 'warped' distance $\mathsf{d}_f(x,y) := f(\|x-y\|)$ is indeed a metric.

For $f \in \mathcal{F}$, define

$$c_b(f) = \inf\left\{\frac{r\cdot\kappa_b(r)\cdot f'(r) - 4\cdot f''(r)}{f(r)} : r > 0\right\}.$$

We show first that this provides an estimate of the coarse Ricci curvature of our diffusion process with respect to the base metric d_f, following the reflection coupling.

Lemma 1. *Let (X_t, Y_t) be the reflection coupling, made absorbing after its coupling time, and define $R_t = \|X_t - Y_t\|$. Before the coupling time, it holds that*

$$\mathrm{d}R_t = \left\langle b(X_t) - b(Y_t), \frac{X_t - Y_t}{R_t}\right\rangle \mathrm{d}t + 2\sqrt{2}\,\mathrm{d}\beta_t.$$

Consequently, for every $f \in \mathcal{F}$, it holds for $\tilde{f} : (x,y) \mapsto f(\|x-y\|)$

$$\left(\widetilde{L}\tilde{f}\right)(x,y) \leq 4\cdot f''(r) - r\cdot\kappa_b(r)\cdot f'(r), \qquad r = \|x-y\|.$$

If $c_b(f) > 0$, then it follows that for all $t \geq 0$,

$$\mathbf{E}\left[f(\|X_t - Y_t\|)\right] \leq \exp(-c_b(f)\cdot t)\cdot f(\|x-y\|).$$

In particular, writing $\mathcal{T}_f$ for the transportation cost on $\mathcal{P}\left(\mathbf{R}^d\right)$ induced by the cost $c(x,y) = f(\|x-y\|)$, it holds that

$$\mathcal{T}_f(\mu P_t, \nu P_t) \leq \exp(-c_b(f)\cdot t)\cdot\mathcal{T}_f(\mu,\nu).$$

Proof. The SDE for R_t is the usual radial equation for reflection coupling. By the definition of κ_b, we bound $\left\langle b(X_t) - b(Y_t), \frac{X_t - Y_t}{R_t}\right\rangle \leq -R_t\cdot\kappa_b(R_t)$. Application of Itô's formula then yields the second claim; the definition of $c_b(f)$ then implies that $\exp(c_b(f)\cdot t)\cdot f(\|X_t - Y_t\|)$ is a local supermartingale; after localization, Grönwall's inequality yields the third claim. Mixing the pointwise reflection couplings over an initial coupling of μ and ν and then minimizing over that coupling yields the fourth and final claim. $\square$

2.1. The constant-curvature-floor construction.

The question then remains of how to design well an f for which $c_b(f)$ is as large as possible. A simplifying strategy is as follows: for 'typical' separation-curvature profiles of interest, $\kappa_b(r)$ is bounded uniformly away from 0 for large r, and so the original metric already works sufficiently well. This suggests that it is reasonable to imagine that for well-chosen f, one will have f' asymptotically constant, and $f(r) \approx r\cdot f'(r)$. From this perspective, one can decompose the expression defining $c_b(f)$ as

$$\frac{r\cdot\kappa_b(r)\cdot f'(r) - 4\cdot f''(r)}{f(r)} = \frac{r\cdot\kappa_b(r)\cdot f'(r) - 4\cdot f''(r)}{r\cdot f'(r)}\cdot\frac{r\cdot f'(r)}{f(r)},$$

and then seek to lower-bound each of the terms in this product separately.

We hence take a parametric approach. We will take $\theta > 0$, construct an explicit$f = f_{b,\theta}$ such that the first factor is at least θ, and control the size of the second term thereafter. With this in mind, define $u_{b,\theta}(r) = \frac{1}{4}\cdot r\cdot(\theta - \kappa_b(r))_+$ and $D_b(\theta) = \int_0^\infty u_{b,\theta}(r)\,\mathrm{d}r = \frac{1}{4}\cdot\int_0^\infty r\cdot(\theta - \kappa_b(r))_+\,\mathrm{d}r$.

Proposition 2. *Suppose that $D_b(\theta) < \infty$. Define*

$$f_{b,\theta}(r) = \int_0^r \exp\left(-\int_0^s u_{b,\theta}(t)\,\mathrm{d}t\right)\mathrm{d}s.$$

Then $f_{b,\theta} \in \mathcal{F}$, and

$$\exp(-D_b(\theta))\cdot r \leq f_{b,\theta}(r) \leq r.$$

Furthermore, there holds the differential inequality

$$4 \cdot f_{b,\theta}''(r) - r \cdot \kappa_b(r) \cdot f_{b,\theta}'(r) \le -\theta \cdot \exp(-D_b(\theta)) \cdot f_{b,\theta}(r),$$

Hence $f_{b,\theta}(\|X_t - Y_t\|)$ contracts with prefactor one at the certified rate $\underline{c}_b(\theta) = \theta \cdot \exp(-D_b(\theta))$.

Proof. By differentiation,

$$f_{b,\theta}'(r) = \exp\left(-\int_0^r u_{b,\theta}(s)\,\mathrm{d}s\right),$$
$$f_{b,\theta}''(r) = -u_{b,\theta}(r) \cdot f_{b,\theta}'(r),$$

so that $f_{b,\theta}$ is increasing and concave. Since $0 \le \int_0^r u_{b,\theta}(s)\,\mathrm{d}s \le D_b(\theta)$, it follows that $\exp(-D_b(\theta)) \le f_{b,\theta}'(r) \le 1$; the bound on $f_{b,\theta}$ then follows by integration.

Moreover,

$$\kappa_b(r) + \frac{4}{r} \cdot u_{b,\theta}(r) = \max\{\kappa_b(r), \theta\} \ge \theta,$$

and therefore

$$4 \cdot f_{b,\theta}''(r) - r \cdot \kappa_b(r) \cdot f_{b,\theta}'(r) = -r \cdot \left(\kappa_b(r) + \frac{4}{r} \cdot u_{b,\theta}(r)\right) \cdot f_{b,\theta}'(r)$$
$$\le -\theta \cdot r \cdot f_{b,\theta}'(r).$$

Finally, observe that since $f_{b,\theta}(r) \le r$ and $f_{b,\theta}'(r) \ge \exp(-D_b(\theta))$, it follows that $r \cdot f_{b,\theta}'(r) \ge \exp(-D_b(\theta)) \cdot f_{b,\theta}(r)$, and hence the claimed differential inequality follows. ☐

Proposition 3. *Define $z_{b,\theta}(r) = \frac{r \cdot f_{b,\theta}'(r)}{f_{b,\theta}(r)}$. The best generator rate associated with the particular metric $f_{b,\theta}$ is*

$$c_b^{\mathrm{ex}}(\theta) = \inf\{\max\{\kappa_b(r), \theta\} \cdot z_{b,\theta}(r) : r > 0\}.$$

In particular, $c_b^{\mathrm{ex}}(\theta) \ge \theta \cdot \exp(-D_b(\theta))$.

Proof. Compute explicitly that

$$-\frac{4 \cdot f_{b,\theta}''(r) - r \cdot \kappa_b(r) \cdot f_{b,\theta}'(r)}{f_{b,\theta}(r)} = z_{b,\theta}(r) \cdot \left(\kappa_b(r) + \frac{4}{r} \cdot u_{b,\theta}(r)\right).$$

The second factor equals $\max\{\kappa_b(r), \theta\}$. Taking the infimum over r yields the first claim; Proposition 2 gives the second. ☐

In this regard, the lower bound $\underline{c}_b(\theta)$ treats the two factors separately: it bounds the effective curvature below by θ, and the distortion factor below by $\exp(-D_b(\theta))$.

Proposition 4. *Let $\Theta_b = \{\theta > 0 : D_b(\theta) < \infty\}$. On every interval contained in Θ_b, the function $\theta \longmapsto \log \underline{c}_b(\theta) = \log\theta - D_b(\theta)$ is strictly concave. Consequently, $\underline{c}_b$ has at most one maximizer on such an interval.*

The one-sided derivatives of D_b are expressible as

$$D_{b,-}'(\theta) = \frac{1}{4} \cdot \int_0^\infty r \cdot \mathbf{1}_{\{\kappa_b(r) < \theta\}}\,\mathrm{d}r$$
$$D_{b,+}'(\theta) = \frac{1}{4} \cdot \int_0^\infty r \cdot \mathbf{1}_{\{\kappa_b(r) \le \theta\}}\,\mathrm{d}r,$$

and hence an interior maximizer $\theta_b^\star$ is characterized by

$$\int_0^\infty r \cdot \mathbf{1}_{\left\{\kappa_b(r) < \theta_b^\star\right\}}\,\mathrm{d}r \le \frac{4}{\theta_b^\star} \le \int_0^\infty r \cdot \mathbf{1}_{\left\{\kappa_b(r) \le \theta_b^\star\right\}}\,\mathrm{d}r.$$

If the level set $\{r : \kappa_b(r) = \theta_b^\star\}$ has zero measure with respect to $r\,\mathrm{d}r$, this reduces to the equation

$$\frac{4}{\theta_b^\star} = \int_0^\infty r \cdot \mathbf{1}_{\left\{\kappa_b(r) < \theta_b^\star\right\}}\,\mathrm{d}r.$$

Proof. The function D_b is convex and nondecreasing because it is an integral of convex nondecreasing functions of θ. Hence $\log\theta - D_b(\theta)$ is strictly concave. The one-sided derivative formulas follow from the corresponding derivatives of $(\theta - a)_+$. At an interior maximum, zero must belong to the superdifferential of $\log\theta - D_b(\theta)$, giving

$$D_{b,-}'(\theta_b^\star) \le \frac{1}{\theta_b^\star} \le D_{b,+}'(\theta_b^\star),$$

and multiplication by four gives the main claim. If $\underline{c}_b$ tends to zero at both endpoints of Θ_b, then the unique maximizer characterized above exists; otherwise the supremum may occur at an endpoint of Θ_b. ☐

The construction above deliberately separates two effects. If

$$u_f(r) = -\frac{f''(r)}{f'(r)}, \qquad z_f(r) = \frac{r \cdot f'(r)}{f(r)},$$

then the pointwise contraction rate generated by f factors as

$$-\frac{4 \cdot f''(r) - r \cdot \kappa_b(r) \cdot f'(r)}{f(r)} = z_f(r) \cdot \left(\kappa_b(r) + \frac{4}{r} \cdot u_f(r)\right)$$

The constant-floor ansatz first forces the second factor to be at least a fixed number θ, and then controls the loss in z_f by the total amount of concavity used. This decoupling is what makes the construction explicit, but it can be conservative: the amount of concavity required at radius r should depend not only on $\kappa_b(r)$, but also on the distortion $z_f(r)$ already accumulated at earlier radii.

This observation suggests treating u_f as a nonnegative control and z_f as its state variable. Spending more control improves the instantaneous effective curvature, but it irreversibly lowers the future value of z_f. For a proposed rate c, the natural feedback is therefore the smallest control that makes the pointwise constraint hold. This turns the metric-design problem into a scalar viability problem in the radial variable. Equivalently, the logarithmic derivative $w = f'/f$ satisfies a Riccati equation with an obstacle encoding concavity.

The value of this reformulation is partly conceptual: it organizes the search for improved rates within the class of increasing concave radial metrics and makes clear where the explicit constant-floor construction may lose information. It is also constructive. Simple curvature minorants lead to piecewise analytic feedback laws, of which the one-switch metric below is the most elementary example.

2.2. **Control and obstacle–Riccati formulation.**

Lemma 5. *Let $f \in \mathcal{F}$, and define $u_f(r) = \frac{-f''(r)}{f'(r)}$, $z_f(r) = \frac{r \cdot f'(r)}{f(r)}$. Then $u_f(r) \geq 0$, $0 < z_f(r) \leq 1$, $z_f(0^+) = 1$, and*

$$z_f'(r) = \frac{1}{r} \cdot z_f(r) \cdot (1 - z_f(r)) - u_f(r) \cdot z_f(r).$$

Furthermore,

$$-\frac{4 \cdot f''(r) - r \cdot \kappa_b(r) \cdot f'(r)}{f(r)} = z_f(r) \cdot \left\{\kappa_b(r) + \frac{4}{r} \cdot u_f(r)\right\}.$$

Proof. Concavity gives $u_f \geq 0$. Since $f(0) = 0$ and f is concave, $f(r) \geq r \cdot f'(r)$, and so $0 < z_f(r) \leq 1$. The normalization at zero gives $z_f(0^+) = 1$. Differentiating z_f yields that

$$z_f'(r) = \frac{f'(r)}{f(r)} + \frac{r \cdot f''(r)}{f(r)} - r \cdot \left(\frac{f'(r)}{f(r)}\right)^2,$$

and substitution then gives the first display. The second display then follows from the definitions of u_f and z_f. □

For a proposed rate $c > 0$ and a current state $z > 0$, the pointwise smallest concavity control compatible with the generator requirement in the preceding lemma is$u_{b,c}(r, z) = \frac{1}{4} \cdot r \cdot \left(\frac{c}{z} - \kappa_b(r)\right)_+$. Whenever the corresponding state trajectory remains positive, this leads to the closed-loop equation

$$z_{b,c}'(r) = \frac{1}{r} \cdot z_{b,c}(r) \cdot (1 - z_{b,c}(r)) - \frac{1}{4} \cdot r \cdot (c - \kappa_b(r) \cdot z_{b,c}(r))_+, \qquad z_{b,c}\left(0^+\right) = 1.$$

Proposition 6. *Fix $c > 0$. Suppose that there exists a positive solution $z_{b,c}$ on $(0, \infty)$ of*

$$z_{b,c}'(r) = \frac{1}{r} \cdot z_{b,c}(r) \cdot (1 - z_{b,c}(r)) - \frac{1}{4} \cdot r \cdot (c - \kappa_b(r) \cdot z_{b,c}(r))_+,$$

such that $z_{b,c}(r) = 1 + \mathcal{O}(r^2)$ as $r \downarrow 0$. Define

$$f_{b,c}(r) = r \cdot \exp\left(\int_0^r \frac{z_{b,c}(s) - 1}{s} \, \mathrm{d}s\right).$$

Then $f_{b,c} \in \mathcal{F}$ and

$$4 \cdot f_{b,c}''(r) - r \cdot \kappa_b(r) \cdot f_{b,c}'(r) \leq -c \cdot f_{b,c}(r).$$

Proof. The assumed behaviour of $z_{b,c}$ at zero makes the defining integral finite and gives $f_{b,c}(0) = 0$, $f_{b,c}'(0) = 1$. Moreover, $\frac{f_{b,c}'(r)}{f_{b,c}(r)} = \frac{z_{b,c}(r)}{r}$, so $f_{b,c}' > 0$. Define $u_{b,c}(r) = \frac{1}{4} \cdot r \cdot \left(\frac{c}{z_{b,c}(r)} - \kappa_b(r)\right)_+$. Using the differential equation for $z_{b,c}$, we obtain

$$\frac{f_{b,c}''(r)}{f_{b,c}'(r)} = \frac{f_{b,c}'(r)}{f_{b,c}(r)} + \frac{z_{b,c}'(r)}{z_{b,c}(r)} - \frac{1}{r} = -u_{b,c}(r).$$

Consequently, $f_{b,c}'' \le 0$, and hence $f_{b,c} \in \mathcal{F}$. Finally,

$$
\begin{aligned}
-\frac{4 \cdot f_{b,c}''(r) - r \cdot \kappa_b(r) \cdot f_{b,c}'(r)}{f_{b,c}(r)} &= z_{b,c}(r) \cdot \left(\kappa_b(r) + \frac{4}{r} \cdot u_{b,c}(r)\right) \\
&= \max\bigl\{\kappa_b(r) \cdot z_{b,c}(r), c\bigr\} \ge c.
\end{aligned}
$$

This proves the claimed differential inequality. □

Under the hypotheses of the preceding proposition, define $w_{b,c}(r) = \frac{f_{b,c}'(r)}{f_{b,c}(r)} = \frac{z_{b,c}(r)}{r}$. Since $f_{b,c}''/f_{b,c} = w_{b,c}' + w_{b,c}^2$, the minimal-feedback equation can equivalently be written as

$$4 \cdot \left(w_{b,c}'(r) + w_{b,c}(r)^2\right) = \min\left\{r \cdot \kappa_b(r) \cdot w_{b,c}(r) - c,\, 0\right\},$$

with $r \cdot w_{b,c}(r) \to 1$ as $r \downarrow 0$. Thus, on the active set $r \cdot \kappa_b(r) \cdot w_{b,c}(r) < c$, the metric spends concavity and satisfies the eigenvalue equation with equality. On the inactive set $r \cdot \kappa_b(r) \cdot w_{b,c}(r) \ge c$, the drift alone supplies the desired contraction rate and $f_{b,c}'' = 0$. This active–inactive decomposition is the obstacle, or free-boundary, aspect of the construction.

We shall not attempt to solve this free-boundary problem for a general curvature profile. Instead, we use the active–inactive decomposition as a design principle. Suppose that the curvature is bounded below by $-L_b$ on an inner region $(0, R)$ and by $K_b > 0$ on the outer region $[R, \infty)$. In the inner region, we enforce the eigenvalue equation against the worst-case curvature, solving

$$4 \cdot F''(r) + L_b \cdot r \cdot F'(r) + c \cdot F(r) = 0.$$

At R, we switch off the concavity control and continue the metric linearly. It then remains only to verify that the positive outer curvature is strong enough for this linear branch to belong to the inactive region. This gives the matching condition in the following theorem.

2.3. **An analytic one-switch construction.**

Theorem 7. *Suppose that, for constants $L_b \ge 0$, $K_b > 0$, and $R > 0$, it holds that*

$$\kappa_b(r) \ge \begin{cases} -L_b & 0 < r < R \\ K_b & r \ge R. \end{cases}$$

For $c > 0$, let $F_{b,c}$ solve

$$4 \cdot F_{b,c}''(r) + L_b \cdot r \cdot F_{b,c}'(r) + c \cdot F_{b,c}(r) = 0$$

with $F_{b,c}(0) = 0$, $F_{b,c}'(0) = 1$. Assume that $F_{b,c} > 0$ on $(0, R]$, that $F_{b,c}' > 0$ on $[0, R]$, and that $K_b \cdot \frac{R \cdot F_{b,c}'(R)}{F_{b,c}(R)} \ge c$.

Define

$$f_{b,c}^{\mathrm{sw}}(r) = \begin{cases} F_{b,c}(r), & 0 \le r \le R, \\ F_{b,c}(R) + F_{b,c}'(R) \cdot (r - R), & r \ge R. \end{cases}$$

Then $f_{b,c}^{\mathrm{sw}} \in \mathcal{F}$ and

$$4 \cdot \left(f_{b,c}^{\mathrm{sw}}\right)''(r) - r \cdot \kappa_b(r) \cdot \left(f_{b,c}^{\mathrm{sw}}\right)'(r) \le -c \cdot f_{b,c}^{\mathrm{sw}}(r).$$

Proof. On $(0, R)$, check that $F_{b,c}''(r) = -\frac{1}{4} \cdot \Bigl(L_b \cdot r \cdot F_{b,c}'(r) + c \cdot F_{b,c}(r)\Bigr) < 0$, so that the inner solution is concave. Since $\kappa_b \ge -L_b$, check that

$$4 \cdot F_{b,c}''(r) - r \cdot \kappa_b(r) \cdot F_{b,c}'(r) \le 4 \cdot F_{b,c}''(r) + L_b \cdot r \cdot F_{b,c}'(r) = -c \cdot F_{b,c}(r).$$

The continuation in the display is linear and matches both the value and derivative at R. Since $F_{b,c}$ is concave and vanishes at zero, it follows that $F_{b,c}(R) \ge R \cdot F_{b,c}'(R)$, and hence that $z(r) = \frac{r \cdot \left(f_{b,c}^{\mathrm{sw}}\right)'(r)}{f_{b,c}^{\mathrm{sw}}(r)}$ is nondecreasing on $[R, \infty)$, and thereby that $z(r) \ge z(R) = \frac{R \cdot F_{b,c}'(R)}{F_{b,c}(R)}$. Using $\kappa_b \ge K_b$, check then that

$$4\left(f_{b,c}^{\mathrm{sw}}\right)''(r) - r\kappa_b(r)\left(f_{b,c}^{\mathrm{sw}}\right)'(r) \le -K_b r \left(f_{b,c}^{\mathrm{sw}}\right)'(r) = -K_b z(r) f_{b,c}^{\mathrm{sw}}(r) \le -c f_{b,c}^{\mathrm{sw}}(r),$$

which proves the final claim. □

Corollary 8. *Suppose $L_b = 0$. Then $F_{b,c}(r) = \frac{2}{\sqrt{c}} \cdot \sin\left(\frac{\sqrt{c}}{2} \cdot r\right)$. Set $a = \frac{1}{2} \cdot \sqrt{c} \cdot R$. The positivity conditions in Theorem 7 hold when $0 < a < \pi/2$, and $\frac{R \cdot F'_{b,c}(R)}{F_{b,c}(R)} = a \cdot \cot a$, and the tail condition on K_b is thus equivalent to $a \cdot \tan a \leq \frac{1}{4} \cdot K_b \cdot R^2$.*

Let $a_b^\star \in \left(0, \frac{\pi}{2}\right)$ be the unique solution of $a_b^\star \cdot \tan a_b^\star = \frac{1}{4} \cdot K_b \cdot R^2$. Then the largest rate within the one-switch ansatz is equal to $c_{\mathrm{sw},b} = \frac{4}{R^2} \cdot \left(a_b^\star\right)^2$, and an associated metric is

$$f_b^{\mathrm{sw}}(r) = \begin{cases} \dfrac{R}{a_b^\star} \cdot \sin\left(\dfrac{a_b^\star}{R} \cdot r\right), & 0 \leq r \leq R, \\ \dfrac{R}{a_b^\star} \cdot \sin\left(a_b^\star\right) + \cos\left(a_b^\star\right) \cdot (r - R) & r \geq R. \end{cases}$$

As $K_b \cdot R^2 \to \infty$, one has that $a_b^\star \to \frac{\pi}{2}$, $c_{\mathrm{sw},b} \to \frac{\pi^2}{R^2}$.

Proof. The first formula follows by solving $4 \cdot F'' + c \cdot F = 0$ with the given initial conditions; the subsequent equations follow by direct calculation. The function $a \mapsto a \cdot \tan a$ is strictly increasing from zero to infinity on $\left(0, \frac{\pi}{2}\right)$, ensuring existence and uniqueness of a solution to the given equation. Maximizing $c = 4 \cdot \frac{a^2}{R^2}$ subject to the tail constraint thus yields the solution. ☐

Corollary 9. *Consider the idealized curvature profile*

$$\kappa_b(r) \geq \begin{cases} 0 & 0 < r < R \\ K_b & r \geq R. \end{cases}$$

For $0 < \theta \leq K_b$, the constant-floor construction gives $D_b(\theta) \leq \frac{1}{8} \cdot \theta \cdot R^2$ and therefore $\underline{c}_b(\theta) \geq \theta \cdot \exp\left(-\frac{1}{8} \cdot \theta \cdot R^2\right)$. If $K_b \geq 8/R^2$, then optimization in θ gives

$$\sup\left\{\theta \cdot \exp\left(-\frac{1}{8} \cdot \theta \cdot R^2\right) : 0 < \theta \leq K_b\right\} = \frac{8}{e} \cdot R^{-2}.$$

By contrast, the one-switch construction satisfies $c_{\mathrm{sw},b} \longrightarrow \frac{\pi^2}{R^2}$ as $K_b \cdot R^2 \to \infty$. As such, in the strong-tail-curvature regime, the one-switch ansatz suggested by the obstacle–Riccati formulation improves the certified constant from $8/e$ to π^2.